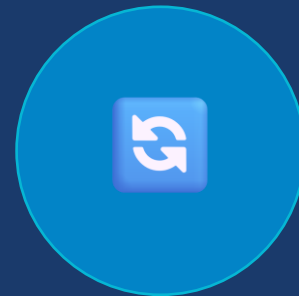


Customer Retention Engine

ML-Powered Churn Prediction & Intervention System

Team Members

Nadir Askarov · Valiyaddin Aliyev · Javidan Muradzada · Matin Khalilov



440K+ Customers

4 ML Models

97.9% Accuracy

Problem Overview



Business Challenge

Customer churn threatens revenue. Without early identification, businesses can't act before customers leave, making retention nearly impossible.



Goal

Build a classification model to predict which customers are likely to churn. Then recommend the minimum interventions to retain them.



Impact

Protect company's revenue, reduce churn rate, and prioritize the retention team's efforts with data-driven, customer-level action plans.

Dataset Description

440,832

Records

12

Features

Binary

Target
(Churn 0/1)

Sample Data

	CustomerID	Age	Gender	Tenure	Usage Frequency	Support Calls	Payment Delay	Subscription Type	Contract Length	Total Spend	Last Interaction	Churn
0	2.0	30.0	Female	39.0	14.0	5.0	18.0	Standard	Annual	932.0	17.0	1.0
1	3.0	65.0	Female	49.0	1.0	10.0	8.0	Basic	Monthly	557.0	6.0	1.0
2	4.0	55.0	Female	14.0	4.0	6.0	18.0	Basic	Quarterly	185.0	3.0	1.0
3	5.0	58.0	Male	38.0	21.0	7.0	7.0	Standard	Monthly	396.0	29.0	1.0
4	6.0	23.0	Male	32.0	20.0	5.0	8.0	Basic	Monthly	617.0	20.0	1.0

● CustomerID — identifier

● Age → Age Group

● Gender — categorical

● Tenure — service duration

● Usage Frequency — engagement signal

● Support Calls — satisfaction signal

● Payment Delay — risk indicator

● Subscription Type — plan tier

● Contract Length — billing cycle

● Total Spend — revenue signal

● Last Interaction — recency

● Churn — target label ✓

EDA — Dataset Overview & Class Balance

Understanding the overall shape, size, and churn distribution before any deeper analysis

440,832

Total Records
after cleaning

12

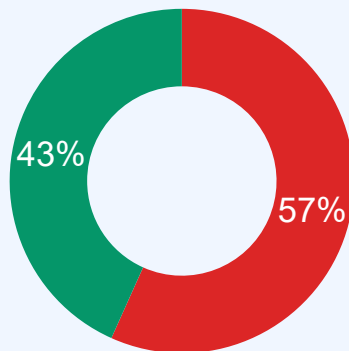
Features
incl. target

250,000

Churn = 1
56.7% of dataset

190,832

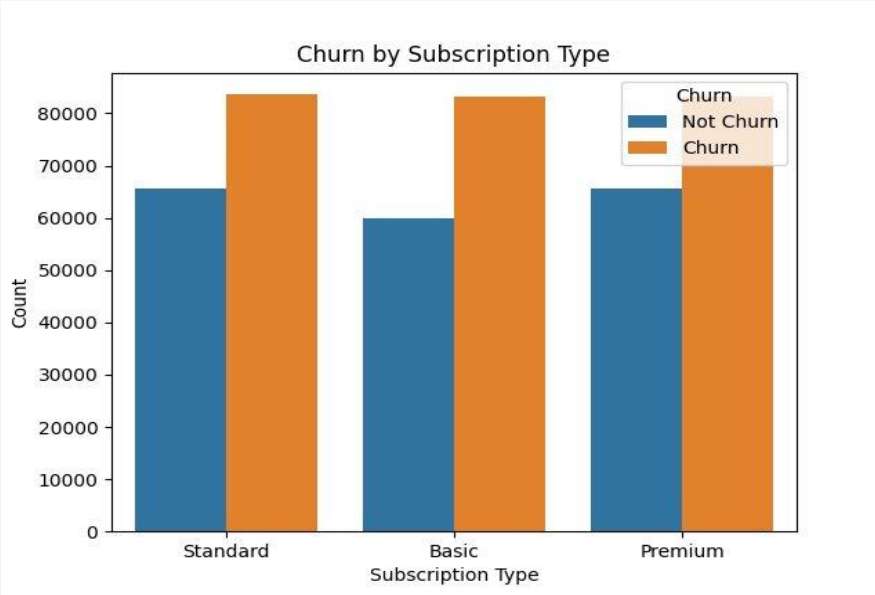
Churn = 0
43.3% of dataset



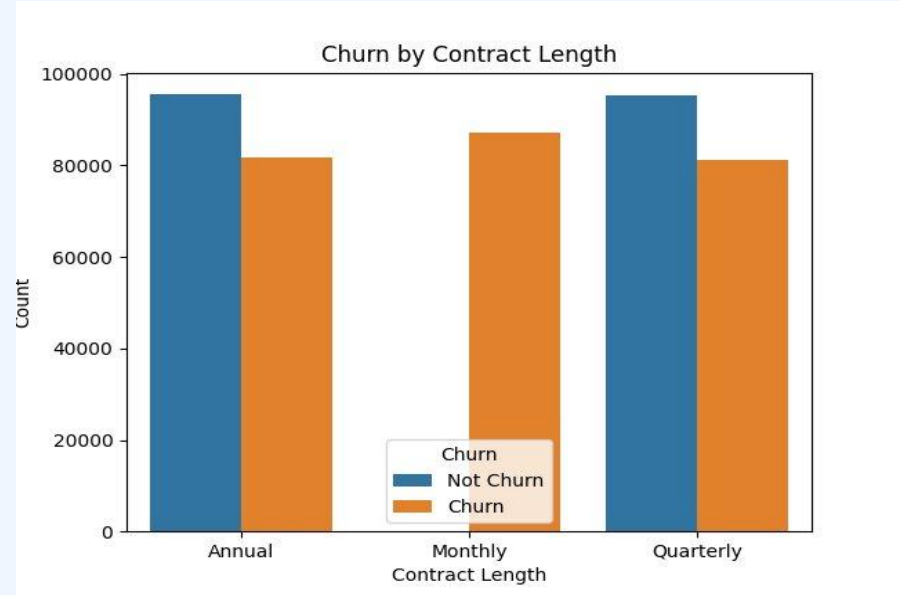
■ Churn (1) ■ Not Churn (0)

EDA — Churn Rates by Subscription & Contract Type

Churn by Subscription Type



Churn by Contract Length



- Standard plan: highest churn (84K churned vs 65K retained)
- Monthly contracts: worst churn-to-retain ratio of all contract types
- Annual contracts: closest to balanced — more stable long-term commitments

EDA — Key Signals & EDA Summary

01



Support Calls

5.1 vs 1.6

Churned customers call support 3× more. High call volume = unresolved frustration.

→ Proactive issue resolution

02



Payment Delay

15.2 vs 10.0 days

Churned customers delay payments by 52% more days on average.

→ Auto-pay enrollment & reminders

03



Total Spend

\$540 vs \$750

Churned customers spend 28% less. Lower spend = lower perceived value

→ Discounts & bundled value offers

04



Contract Length

Monthly > churn

Monthly contracts have the worst churn ratio.

→ Upgrade to longer contract terms

EDA Conclusion: Churn is a behavioral pattern. The top signals — support calls, payment delay, total spend, and contract type — all have direct, actionable business levers, making this dataset ideal for intervention-based retention.

Feature Engineering & Preprocessing



Feature Engineering

AgeGroup derived from Age

Bins: [15,25), [25,35), [35,45), [45,66)

Labels

Child · Youth · Adult · Senior

Raw Age dropped

AgeGroup used in model instead

CustomerID removed

Not a predictive feature



Preprocessing Pipeline

1

Numerical Features

Standard Scaler

Features: Tenure, Usage Frequency, Support Calls,
Payment Delay, Total Spend, Last Interaction

2

Categorical Features

One Hot Encoder

Features: Gender, Subscription Type,
Contract Length, AgeGroup

3

Train / Test Split

80% Train | 20% Test

Test: 88.167 rows

Model Building

All 4 models trained on the same stratified 80/20 split with identical preprocessing — ensuring fair comparison



Logistic Regression

Linear baseline. Tests if churn is linearly separable. Fast, interpretable, but limited for complex patterns.



Decision Tree

Non-linear thresholds and feature interactions. Single-tree.



Random Forest

Bagging ensemble of 100 trees. Reduces variance, improves generalization on tabular data.



Gradient Boosting

Sequential weak learner correction. Focuses on residual errors

Results & Model Comparison

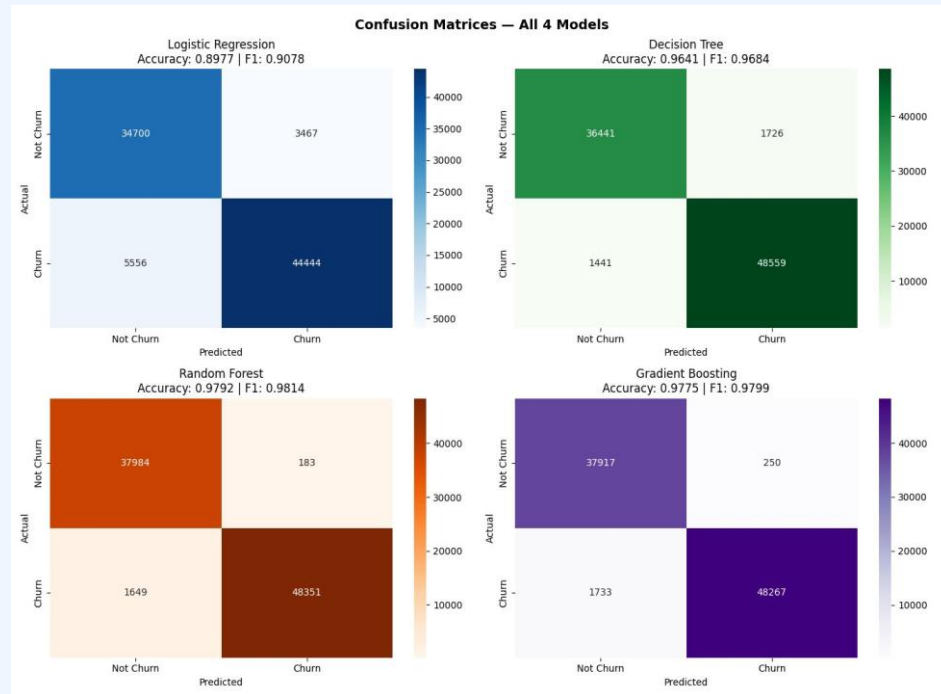
Performance Metrics

Model	Accuracy	F1 Score
1 Random Forest	0.9792	0.9814
2 Gradient Boosting	0.9775	0.9799
3 Decision Tree	0.9641	0.9684
Logistic Regression	0.8977	0.9078

Takeaways

- Ensemble models (RF & GB) drastically cut both false positives and false negatives
- Random Forest leads with F1 0.9814 - only 183 false positives in 88K test rows
- Logistic Regression confirms churn can't be fully captured with linear boundary

Confusion Matrices — All 4 Models



Best Model & Feature Importance



BEST MODEL

Random Forest

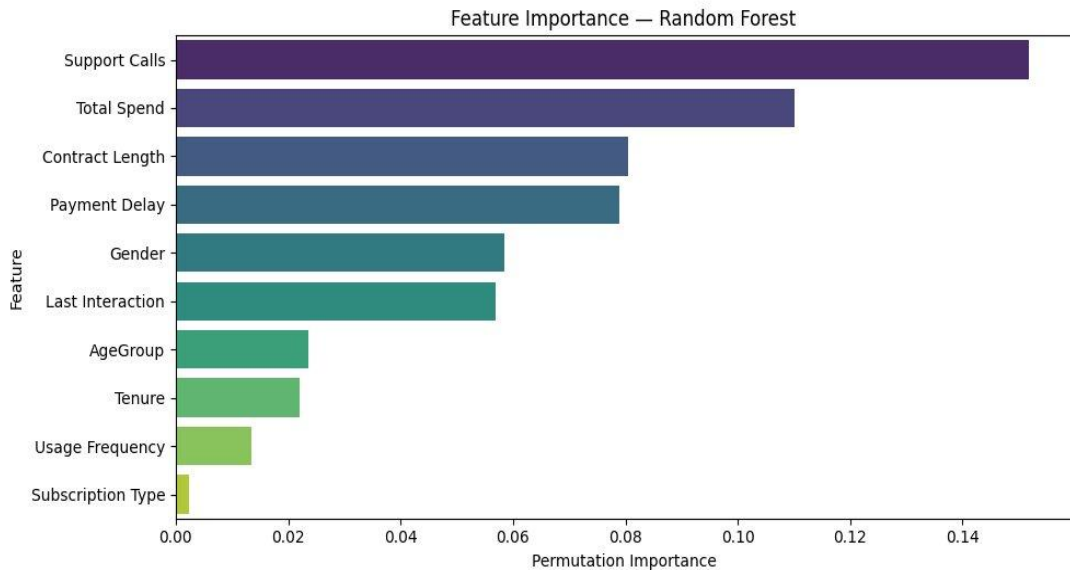
0.9814

F1 Score

0.9792

Accuracy

Feature Importance

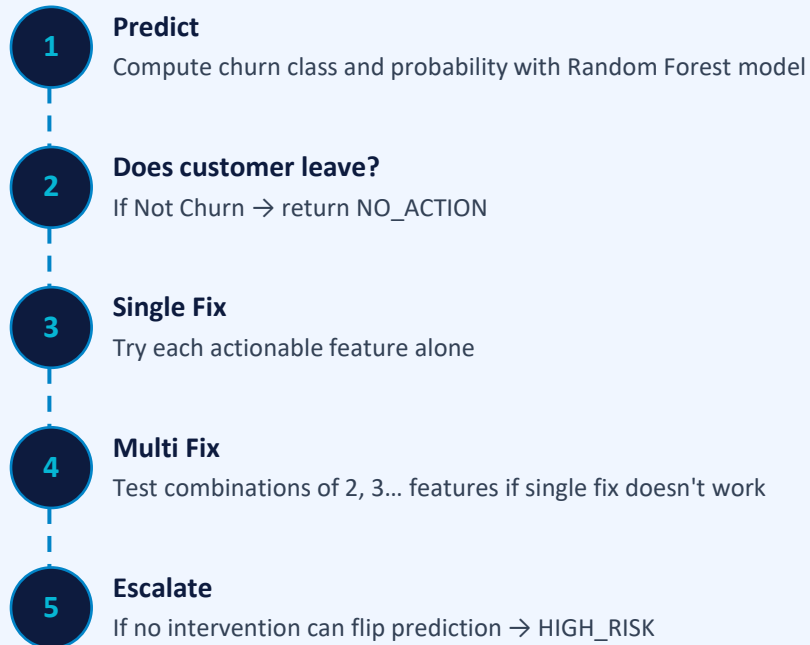


Top Drivers:

1. Support Calls
2. Total Spend
3. Contract Length
4. Payment Delay

Recommendation System

Counterfactual Intervention Engine



Worked Example

Customer Profile

Female · Adult · Premium · Quarterly
Support Calls: 15 · Payment Delay: 29
Total Spend: 233 · Tenure: 27

⚠ **Predicted: CHURN — 100% Probability**

MULTI_CHANGE Interventions:

Total Spend: 233 → 533

Offer discount / bundled value

Payment Delay: 29 → 20 days

Auto-pay enrollment / reminders

Support Calls: 15 → 4 calls

Proactive issue resolution

✓ **New Churn Probability: 16%**

Recommendation System

Feature	Direction & Step	Business Action	Why It Works
Total Spend	↑ Increase +\$50 per iteration	Offer personalised discount, promotional bundle, or loyalty reward to increase spending commitment	<i>Higher spend signals stronger product engagement</i>
Payment Delay	↓ Decrease -1 day per iteration	Send payment reminders, enroll customer in auto-pay, or provide billing assistance	<i>Reducing delay reduces financial disengagement risk.</i>
Support Calls	↓ Decrease -1 call per iteration	Proactively resolve open issues, improve self-service resources, and assign dedicated support contacts	<i>High call volume = unresolved frustration — the #1 predictor.</i>
Contract Length	↔ Change category Upgrade plan	Offer contract upgrade incentives (e.g. discounted annual plan)	<i>Longer commitments create switching cost and signal customer confidence.</i>

Conclusion



Dataset Description

Customer churn dataset contains user's behavior, pattern, activity and retention labels



EDA Insights

Support Calls, Total Spend, Contract Length and Payment Delay are strongest behavioral signals



Best Model: Random Forest

F1 = 0.9814 | Accuracy = 0.9792



Recommendation Engine

Recommends minimal customer-level interventions that can flip a predicted outcome to non-churn



Business Value

Preventing customer churn protects revenue and allows retention teams focus on data-driven actions.

Thank You

Questions?